
Bring Back the Bladderpod
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Bladderpod: systematics and ecology of *Physaria globosa* [Brassicaceae].
Notes by Julian Campbell, Jun 2014; http://bluegrasswoodland.com/Notes_on_Species.html

In a preliminary trial, I have grown bladderpod from several provenances in my garden for the past few years; following are general locations for provenances (with original discoverer in parentheses).

Kentucky: these are becoming mixed but started with just Jessamine County.

Franklin County: rocky woods along Little Benson Creek (T. Littlefield find).

Jessamine County: roadside north of Wilmore (J. Brushaber find in 1990s).

Woodford County: rocky yard west of Versailles (see KSNPC database)

Tennessee

Davidson County: rocky woods along Cumberland Rv. west of Nashville (D. Estes).

Indiana

Posey County: edge of gravel road on bottomland of Wabash Rv (M. Homoya).

I have been keeping TN and IN provenances separate by at least 100 ft, to minimize cross-pollination, but I also allowed some plants in the KY and TN populations to cross in 2012 by placing some flats together. There is some variation in overall robustness, and I was curious to see if some 'hybrid vigor'might emerge. For KY and TN plants, I grew the pure KY, pure TN plants and potentially mixed groups out in two sets of plots: (1) a relatively shady site; (2) a relatively sunny site. It appeared that leaf size might be a parameter for initial investigation, and I took a small sample from the two sites for each provenance plus the mixed KY-TN plants. For the IN plants, I just took samples from the sunny site. In each of these nine batches of plant (4 KY-TN groups x 2 sites plus the IN plants), I selected three of the largest plants for

sampling. I selected two types of leaf for measurement: (a) the largest basal rosette leaf in each plant when flowering began in May 2014; (b) the largest leaf at lowest branch of inflorescence. For each leaf type, I measured length (including petiole) and width at the widest section (including any teeth). Since patterns of variation in length and width appeared to be generally similar, I used their product as a more precise parameter (length × width in square cm).

Differences between some of these groups of plants appeared to be significant, but a more formal experiment is needed for proper analysis. The groups were not randomly mixed, and better replication is needed. Although plants in sun were much larger, there was no consistent difference between sunny and shady sites in overall leaf size. Thus each group had six samples for direct comparison (3 from sun plus 3 from shade), or just 3 in the case of IN.

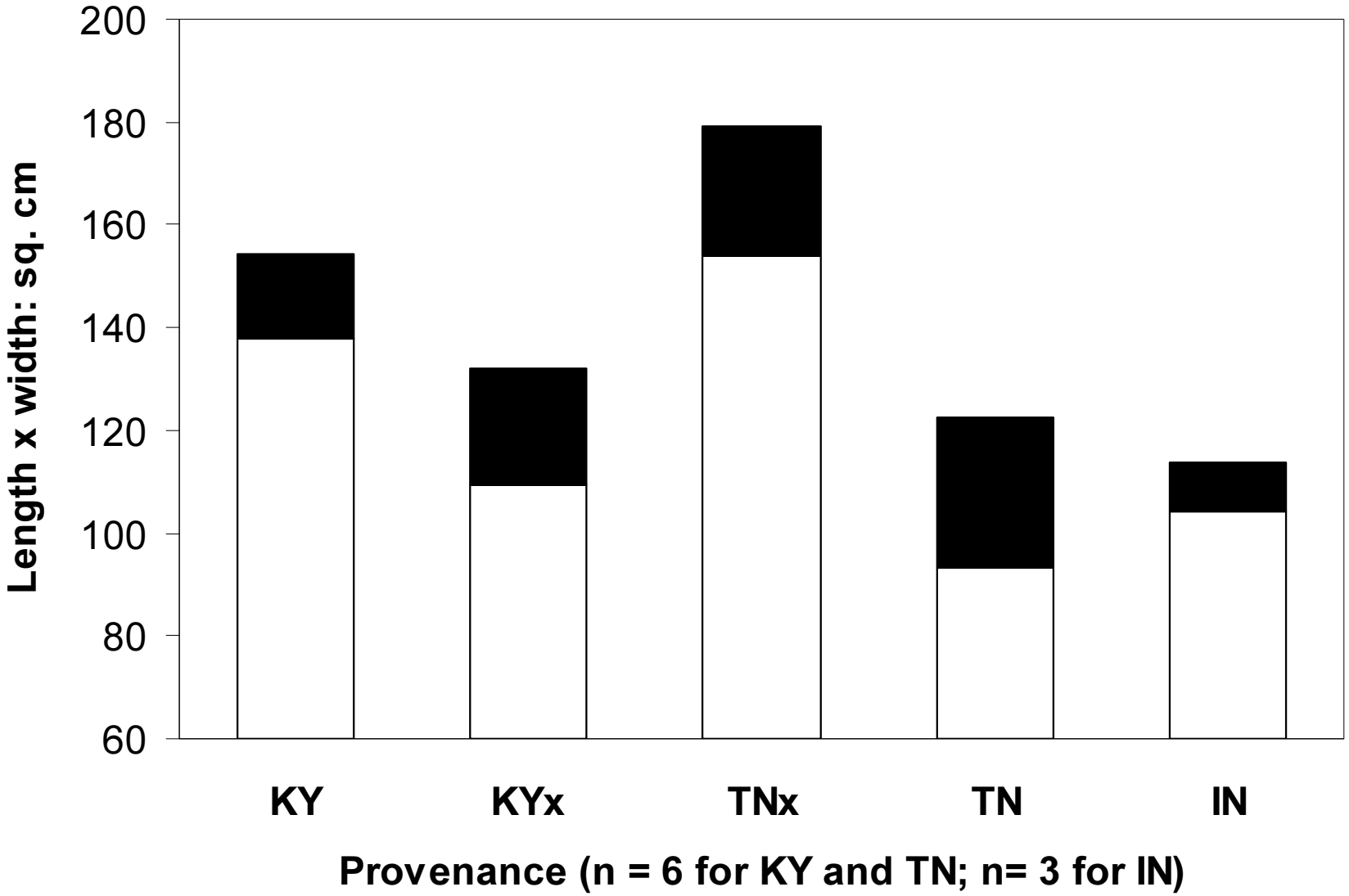
Using ANOVA, there were significant differences among some of these genetic groups ($P<0.05$). However, pure plants from KY, TN and IN were generally similar, except that pure KY plants had larger basal leaves than TN and IN plants (by about 40%). Plants from outcrossed TN mothers had larger basal leaves (by about 50% from pure TN) and larger inflorescence leaves (by about 15%). In contrast, plants from outcrossed KY mothers had smaller basal leaves than pure KY plants (marginally significant) and just slightly larger inflorescence leaves than pure KY plants (not significant).

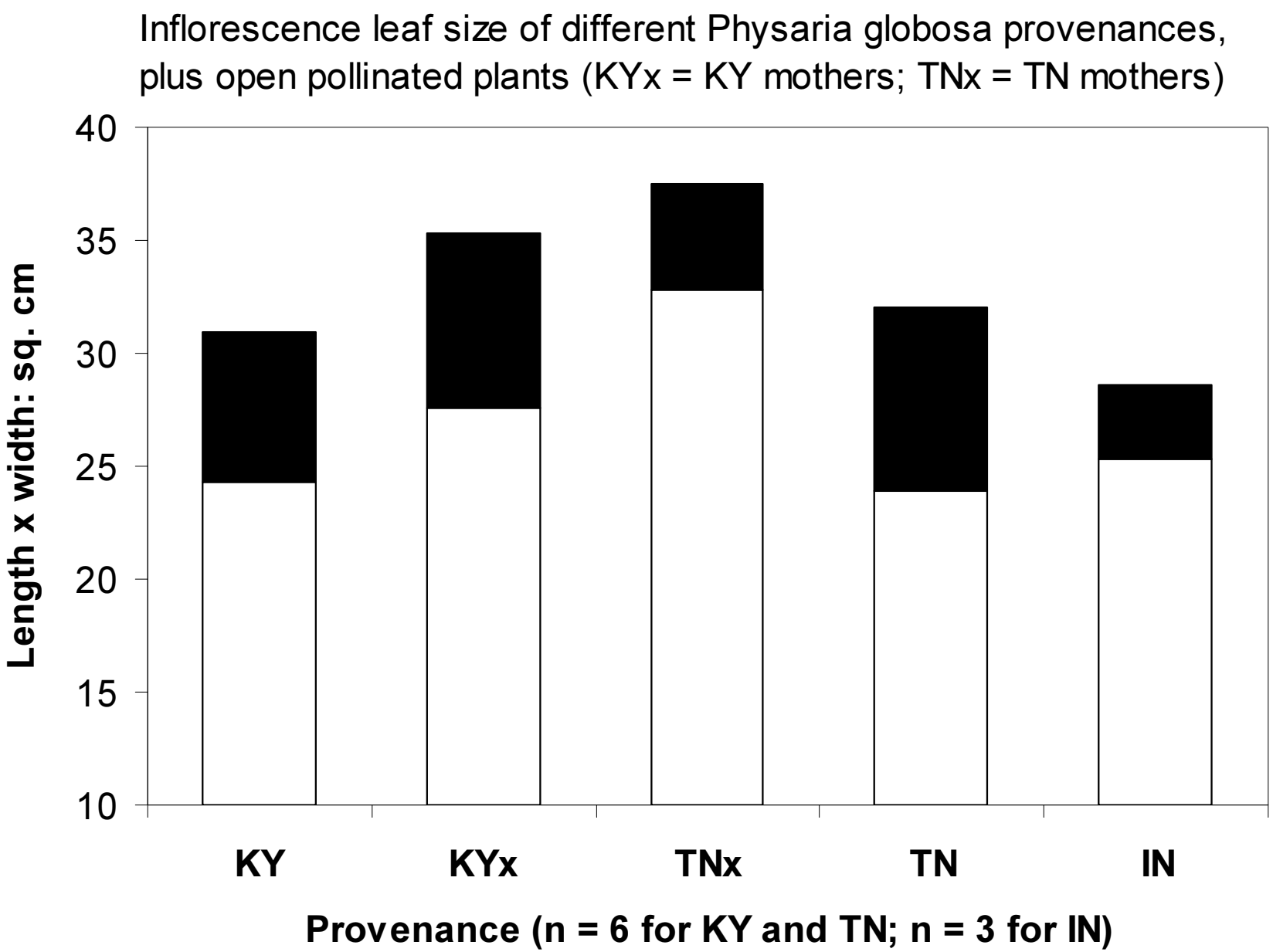
This work would be worth extending into a general recovery program for the species, which probably used to be a much more common weed along trails in the Ohio Valley, especially those leading to and from the central Bluegrass of KY and cedar glade region of TN. I hope to develop more robust aggressive plants that can persist in general suburban environments.

Figure 1a and 1b (next pages).

Leaf sizes of *Physaria globosa* provenances and outcrossed groups. Parameters (y axis) are length × width for largest basal leaves (a) and for leaves at lowest branch of inflorescence (b). Black bars span the means: lower margin is mean minus standard error; upper margin is mean plus standard error. In most cases, overlapping bars are not significantly different at $P < 0.05$.

Basal leaf size of different Physaria globosa provenances, plus open pollinated plants (KYx = KY mothers; TNx = TN mothers)




Inflorescence leaf size of different Physaria globosa provenances, plus open pollinated plants (KYx = KY mothers; TNx = TN mothers)
Length x width: sq. cm
40
35
30
25
20
15
10
KY
KYx
TNx
TN
IN
Provenance (n = 6 for KY and TN; n = 3 for IN)

